urgencies, a useful indication of when the human species is ready to consider terraforming seriously is when we have put our own world right. We can consider it a test of the depth of our understanding and our commitment. The first step in engineering the Solar System is to guarantee the habitability of the Earth.

Then we'll be ready to spread out to asteroids, comets, Mars, the moons of the outer Solar System, and beyond. Jack Williamson's prediction that this will begin to come about by the twenty-second century may not be far off the mark.

THE NOTION OF OUR DESCENDANTS living and working on other worlds, and even moving some of them around for their convenience, seems the most extravagant science fiction. Be realistic, a voice inside my head counsels. But this *is* realistic. We're on the cusp of the technology, near the midpoint between impossible and routine. It's easy to be conflicted about it. If we don't do something awful to ourselves in the interim, in another century terraforming may seem no more impossible than a human-tended space station does today.

I think the experience of living on other worlds is bound to change us. Our descendants, born and raised elsewhere, will naturally begin to owe primary loyalty to the worlds of their birth, whatever affection they retain for the Earth. Their physical needs, their methods of supplying those needs, their technologies, and their social structures will all have to be different.

A blade of grass is a commonplace on Earth; it would be a miracle on Mars. Our descendants on Mars will know the value of a patch of green. And if a blade of grass is priceless, what is the value of a human being? The American revolutionary Tom Paine, in describing his contemporaries, had thoughts along these lines:

> The wants which necessarily accompany the cultivation of a wilderness produced among them a state of society which countries long harassed by the quarrels and intrigues of governments had neglected to cherish. In such a situation man becomes what he ought to be. He sees his species … as kindred.